Experiment 28: Medical Diagnosis Using Prolog

Aim

To develop a **simple rule-based medical diagnosis system using Prolog** that identifies possible diseases based on a set of symptoms.

Objective

1. To represent symptoms and diseases using Prolog facts and rules.
2. To understand knowledge representation in Prolog.
3. To use logical rules to identify possible diseases.
4. To execute queries and obtain the corresponding diagnosis.

Algorithm

1. Start the program.
2. Define the symptoms as Prolog facts.
3. Define rules connecting combinations of symptoms with possible diseases.
4. Enter a query for a disease or diagnosis.
5. Prolog checks the required symptoms for that disease.
6. If all required symptoms are available, return true.
7. Otherwise, return false.
8. Use a variable to find all possible diagnoses.
9. Display the result.
10. Stop.

Flowchart

```
          START
            |
      Define symptoms
            |
    Define disease rules
            |
      Enter the query
            |
    Check required symptoms
            |

     Are symptoms present?
        /          \
       YES           NO
        |                  |
 Possible diagnosis    false
        |
  Display diagnosis
        |
      STOP
```

Prolog Code

```
% Medical Diagnosis System

% Symptoms

symptom(fever).
symptom(cough).
symptom(body_pain).
symptom(sore_throat).
symptom(runny_nose).
symptom(headache).
symptom(rash).

% Diagnosis rules
```

```
diagnosis(flu) :-
    symptom(fever),
    symptom(cough),
    symptom(body_pain).

diagnosis(common_cold) :-
    symptom(cough),
    symptom(runny_nose),
    symptom(sore_throat).

diagnosis(measles) :-
    symptom(fever),
    symptom(rash),
    symptom(headache).

% Rule to display diagnosis

medical_diagnosis(Disease) :-
    diagnosis(Disease).
```

Sample Output

Query 1: Check for Flu

```
?- diagnosis(flu).
```

Query 2: Check for Common Cold

```
?- diagnosis(common_cold).
```

Query 3: Check for Measles

```
?- diagnosis(measles).
```

Query 4: Find all possible diagnoses

```
?- medical_diagnosis(Disease).
```

Query 5: Check an unknown disease

?- diagnosis(malaria).

Result

The **medical diagnosis system was successfully implemented using Prolog**. The program correctly identifies possible diseases based on predefined symptoms and rules.

Conclusion

Thus, the experiment demonstrates how **Prolog can be used to build a simple rule-based medical diagnosis system**. Facts represent symptoms, while logical rules are used to determine possible diseases.

Note: This is an academic demonstration only and should not be used for actual medical diagnosis or treatment.